

Experiment 4

NMOS Common-Source Amplifier with Resistive Load: DC Transfer Characteristic and Small-Signal Gain

Important: V_{TH} , k_n and λ above, and every numerical value in the Section 6 reference plot, are illustrative only, chosen purely to show the shape of the curves and how the two gain-extraction methods are meant to be compared. Your NMOS device (and every device on your bench) will very likely have different V_{TH} , k_n and λ , and a different safe current limit. Do not copy these numbers into your report. Determine your own device's parameters from your Experiment 3 transfer/output-characteristic data (or the datasheet) before predicting or interpreting your Part A and Part B results, and expect your measured V_{OUT} vs V_{IN} curve and gain curve to be shifted and scaled differently from Section 6.

1. Objective

To construct a common-source (CS) NMOS amplifier with a resistive drain load, measure and plot its DC transfer characteristic (V_{OUT} vs V_{IN}), extract the small-signal voltage gain as a function of the DC bias point in two independent ways, and verify that the two agree.

2. Circuit

The gate of an enhancement NMOS is driven by V_{IN} . The drain is connected to V_{DD} through a resistor R_D and is the output node, V_{OUT} . The source is grounded. A DC supply sets V_{IN} through a potentiometer or a programmable source; a small AC signal from a function generator is superposed on V_{IN} through a coupling capacitor and blocking resistor for Part B.

3. Nominal Parameters

Use your own device's parameters: take V_{TH} , k_n (or the equivalent transconductance parameter) and the safe current limit from your Experiment 3 characterization of this specific NMOS. The table below lists the values used only to generate the reference figure in Section 6 - they are not your device's values.

Parameter	Illustrative value used ONLY for the reference plot in Section 6
Supply voltage, V_{DD}	10 V
Drain resistor, R_D	4.7 k Ohm
Threshold voltage, V_{TH}	2.0 V
Transconductance parameter, k_n	3 mA/V ²
Channel-length modulation, λ	0.02 /V

4. Part A - DC Transfer Characteristic

Procedure:

- Sweep V_{IN} from 0 V to V_{DD} in steps of about 0.2 V (finer steps of 0.05 V near your device's V_{TH} and near the saturation-to-triode boundary).
- At each step, record V_{OUT} with the DMM/scope. Do not let I_D exceed your device's datasheet current limit.
- Identify and label the three regions on your plot: cutoff, saturation (the amplifying region), and triode. The V_{IN} at which each region begins depends on your device's own V_{TH} and on R_D and V_{DD} as you've wired them - it will not match Section 6.

Plot V_{OUT} vs V_{IN} on linear axes for the full sweep.

5. Part B - Small-Signal Gain vs. DC Bias Point (Two Methods)

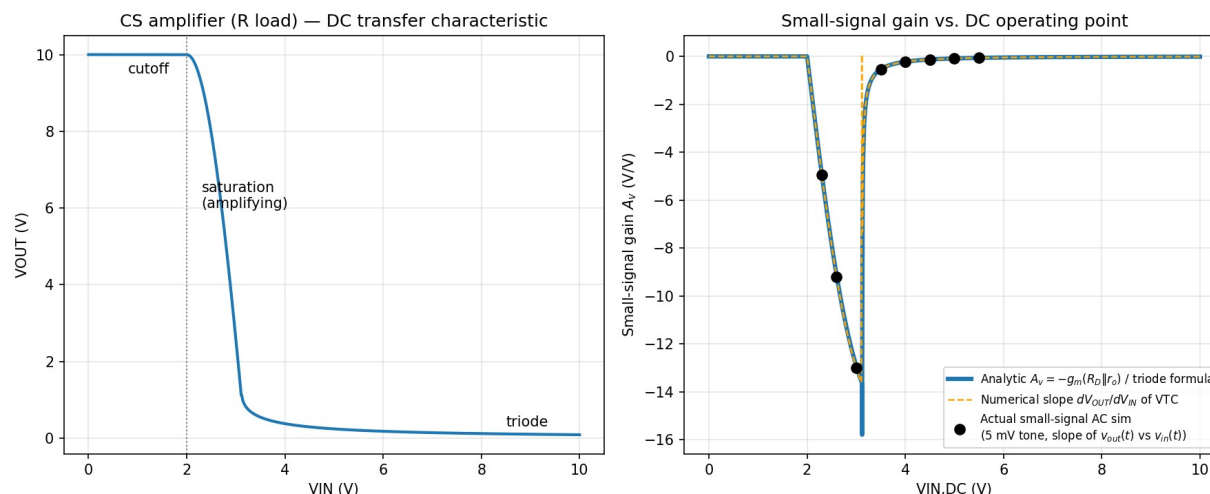
Method 1 - Slope of the measured transfer characteristic: compute $A_v(V_{IN}) = d(V_{OUT})/d(V_{IN})$ as the numerical derivative (point-to-point slope) of the V_{OUT} vs V_{IN} data from Part A. This does not require any additional measurement.

Method 2 - Direct AC measurement: for a chosen set of DC bias points $V_{IN,DC}$ (span cutoff edge, mid-saturation, and near-triode for your own device), superpose a small AC tone (e.g. 5-10 mV amplitude, 1 kHz) on V_{IN} . On the scope, measure the AC output amplitude and divide by the AC input amplitude: $A_v = -(v_{out,ac,pp})/(v_{in,ac,pp})$. Keep the AC amplitude small enough that $v_{out}(t)$ stays sinusoidal (no visible clipping or asymmetry) - this is what makes it a valid small-signal measurement.

Tabulate A_v from both methods at the same set of $V_{IN,DC}$ points and compare them directly.

6. Reference Simulation (illustrative values only)

The figure below solves the same circuit equations using the illustrative parameters of Section 3 - it is included purely to show the expected shape of the curves and how Methods 1 and 2 should line up, not as a numerical prediction for your device. Your own V_{TH} , k_n , λ , V_{DD} and R_D will shift and rescale both the transfer characteristic and the gain curve. Left: DC transfer characteristic with the three regions marked. Right: small-signal gain vs. $V_{IN,DC}$ computed three ways - the analytic derivative of the device equations (solid line), the numerical slope of the simulated transfer characteristic (dashed line), and a simulated small-AC-signal measurement at several bias points (points), mirroring Methods 1 and 2 above.



7. Observation Table (to be filled during the session, using your own device)

V_{IN} (V)	V_{OUT} (V), measured	Region (cutoff / saturation / triode)

8. Analysis and Discussion Questions

- Where along the V_{IN} axis is the gain magnitude largest for your device? Relate this to g_m and to how close V_{OUT} sits to the edge of saturation.
- Why does the gain fall toward zero in cutoff and again deep in triode?
- How closely do your Method 1 and Method 2 gain values agree at each bias point? Account for any discrepancy (measurement noise, AC amplitude not small enough, step size in the DC sweep, etc.).
- If R_D is increased, how would you expect the saturation-region gain and the extent of the triode region to change?
- State, in one or two lines, why the slope of the large-signal DC transfer curve at a bias point equals the small-signal AC gain measured at that same bias point.
- Compare your device's V_{TH} and the V_{IN} at which your gain peaks against the illustrative Section 6 values - by how much, and in which direction, do they differ?

9. Precautions

- Respect YOUR device's datasheet current limit at all times; do not carry over the illustrative limit implied by Section 6's parameters.
- Keep the AC test signal amplitude small (a few mV) so the small-signal approximation remains valid.
- Ensure the coupling capacitor and bias network do not load down $V_{IN,DC}$ when the AC source is connected.